VINUNIVERSITY

Test duration: 1 hour and 30 minutes

Name: **Student ID:**

Midterm Practice Test 1 – Solutions – Fall 2026

COMP3040/5030 Computer Vision Unofficial practice test, prepared by Lê Tăng Phú Quý

Instructions:

- Academic honesty is required in all work you submit to be graded. You should **NOT** copy or share your solutions with other students to avoid plagiarism issues.
- For short answer questions, please make your answers as **concise** as possible.
- This is a closed-book exam and NO Internet access is allowed. You are allowed 1 page of **hand-written** note (both sides).
- Drafting paper and pen/pencil are allowed. We will provide the drafting paper to you.
- Late submission without an approved extension will not be graded at all.
- You must **RETURN** this hard copy of the midterm exam at the end of the exam period.
- During the exam, you are **NOT** allowed to leave the room or/and use the restroom.
- You are required to **NOT** use your mobile phone in the exam room. Please put it on the table or in your bag.
- Good luck!

About the exam: There are **05** problems with **100 points** total, and some have multiple parts. Please check through the pages before you start.

Problem	Content	Point (Undergraduates)	Type
1	Fundamentals of Computer Vision	20	short answer
2	Image Filtering	25	long answer
3	Feature Detection and Matching	25	long answer
4	Fitting and Image Transformations	25	long answer
5	System Design	05	long answer
Total		**100**	

Problem 1. Fundamentals of Computer Vision

Give a short answer for each of the following question:

1.1. What is the difference between image **filtering** and image **warping**? Give one example of each. (5 pts)

Solution. Filtering changes **values** (range): $g(\mathbf{x}) = h(f(\mathbf{x}))$ or a neighbourhood combination — e.g. Gaussian blur, negative. Warping changes **positions** (domain): $g(\mathbf{x}) = f(T(\mathbf{x}))$ — e.g. rotation, scaling, a homography.

1.2. Write the **negative** transformation for an 8-bit grayscale image and for an image normalized to $[0, 1]$. Why is it a *point operator*? (5 pts)

Solution. $g = 255 - f$ and $g = 1 - f$. A point operator because each output depends only on that pixel's own value (no neighbours) — it can be a 256-entry look-up table.

1.3. Name the four main steps of SIFT. (5 pts)

Solution. (1) Scale-space extrema of the DoG; (2) keypoint localization (Taylor fit, reject low contrast & edges); (3) orientation assignment (36-bin histogram); (4) 128-D descriptor (4×4 cells × 8 orientations).

1.4. Why do we need **homogeneous coordinates** to represent affine transformations? (5 pts)

Solution. Translation is not linear (it moves the origin), so no 2×2 matrix can represent it. Append a 1: $(x, y, 1)$, and $\mathbf{p}' = T\mathbf{p} + \mathbf{b}$ becomes **one** 3×3 matrix product; composition = matrix product, inverse = matrix inverse.

Problem 2. Image Filtering

2.1. A 5×5 image has $I[i, j] = (i+1)(j+1)$ (row i, column j, from 0). It is filtered with a 3×3 **box** (mean) filter. (a) Output size with padding 0, stride 1? padding 1, stride 1? padding 1, stride 2? (b) The value at the centre $(2, 2)$. (c) The value at the corner $(0, 0)$ with zero padding and with replicate padding; comment. (12 pts)

Solution. (a) $\lfloor (n + 2p - k)/s \rfloor + 1$: 3×3; 5×5; $\lfloor (5 + 2 - 3)/2 \rfloor + 1 = 3 \Rightarrow 3 \times 3$.

(b) Rows/columns 1–3: sum $= (2 + 3 + 4)(2 + 3 + 4) = 81 \Rightarrow 81/9 = 9$.

(c) Zero padding: only 4 real pixels $\{1, 2, 2, 4\}$, sum 9 $\Rightarrow 9/9 = 1$ (the border **darkens**). Replicate: rows/columns contribute $(1 + 1 + 2)$ each $\Rightarrow 16/9 \approx 1.78$. Zero padding darkens the border; replicate/reflect preserve brightness.

2.2. A 3×3 window holds $\{12, 15, 255, 14, 13, 0, 16, 11, 14\}$. Compute the output of a **mean** filter and of a **median** filter. What kind of noise is this, and which filter should be used? (8 pts)

Solution. Mean $= 350/9 \approx 38.9$ (pulled by the two impulses). Sorted: 0, 11, 12, 13, **14**, 14, 15, 16, 255 $\Rightarrow$ median $= 14$. **Salt-and-pepper** noise (impulses 0 and 255) $\Rightarrow$ use the **median**: an impulse is never the median, and edges stay sharp.

2.3. Is the median filter linear? Prove it or give a counterexample. (5 pts)

Solution. No. With 3 samples: $\mathrm{med}(1,0,0)=0$, $\mathrm{med}(0,0,1)=0$ but $\mathrm{med}(1,0,1)=1\neq 0+0$. (It is still shift-invariant.)

Problem 3. Feature Detection and Matching

3.1. For the patch $\begin{bmatrix}10&10&10\\10&10&50\\10&50&50\end{bmatrix}$ and the (correlation-form) Sobel kernels $S_x=\begin{bmatrix}-1&0&1\\-2&0&2\\-1&0&1\end{bmatrix}$, $S_y=S_x^\top$: (a) compute I_x, I_y at the centre; (b) the gradient magnitude and direction (y axis pointing down); (c) in Canny's NMS, which two neighbours is the centre compared with? (13 pts)

Solution. (a) $I_x=(10-10)+2(50-10)+(50-10)=120$; $I_y=(10-10)+2(50-10)+(50-10)=120$.

(b) $\|\nabla I\|=120\sqrt{2}\approx 169.7$; $\theta=\mathrm{atan2}(120,120)=45^\circ$ — pointing down-right (towards the bright side).

(c) The 45° bin: compare with $(y+1,x+1)$ and $(y-1,x-1)$; keep it if it is $\geq$ both.

3.2. Template matching with SSD, NSSD and ZNCC. (a) Write the formulas. (b) If the source image is re-taken in darker light (every pixel $I\to 0.6I+10$), which measure is still reliable, and why? (c) Why does template matching fail when the object is rotated by 30°? (12 pts)

Solution. (a) SSD $=\sum(T-P)^2$; NSSD $=\sum(T-P)^2/\sqrt{\sum T^2\sum P^2}$; ZNCC $=\frac{\sum(T-\bar T)(P-\bar P)}{\sqrt{\sum(T-\bar T)^2\sum(P-\bar P)^2}}$.

(b) **ZNCC**: subtracting the mean removes $+10$, dividing by the spread removes the factor 0.6 $\Rightarrow$ invariant to $aI+b$, $a>0$. SSD changes a lot; NSSD only reduces sensitivity to the gain, not the offset.

(c) The template is only translated, never rotated: corresponding pixels no longer line up. One must try many angles/scales (costly) or use invariant features (SIFT).

Problem 4. Fitting and Image Transformations

4.1. Fit $y=mx+b$ to $(1,1),(2,3),(3,2),(4,4)$ by least squares. Write the normal equations, find m,b and the sum of squared errors. (13 pts)

Solution. $\sum x=10$, $\sum x^2=30$, $\sum y=10$, $\sum xy=29$, $n=4$: $\begin{bmatrix}30&10\\10&4\end{bmatrix}\begin{bmatrix}m\\b\end{bmatrix}=\begin{bmatrix}29\\10\end{bmatrix}\Rightarrow$ $m=\frac{4\cdot 29-10\cdot 10}{20}=0.8$, $b=\frac{30\cdot 10-10\cdot 29}{20}=0.5$. Residuals $-0.3, 0.9, -0.9, 0.3\Rightarrow E=1.8$.

4.2. (a) Rotate $(2,1)$ by 90° counter-clockwise about the origin (math axes, y up), then translate by $(3,-1)$. Write the 3×3 homogeneous matrix and the result. (b) Redo in the opposite order. What do you conclude? (12 pts)

Solution. (a) $T(3,-1)R_{90}=\begin{bmatrix}0&-1&3\\1&0&-1\\0&0&1\end{bmatrix}$; $(2,1)\to(-1,2)\to(2,1)$. (b) $R_{90}T(3,-1)=$

$\begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix}$; $(2,1) \rightarrow (5,0) \rightarrow (0,5)$. Composition is **not commutative**; the first-applied matrix sits on the right.

Problem 5. System Design

5. You build a **prescription digitizer**: users photograph a paper prescription with a phone (tilted, skewed, cluttered desk). Design an automatic pipeline that turns the photo into a canonical, front-facing, fixed-size image of the page, ready for OCR. (5 pts)

Solution. (1) Grayscale + Gaussian denoising; (2) Canny edges; (3) (ρ, θ) Hough for strong lines, merge duplicate peaks, keep two near-parallel pairs forming the largest quadrilateral; (4) intersect line pairs $\Rightarrow$ the 4 page corners, ordered TL, TR, BR, BL; (5) estimate the homography from the 4 corners to a canonical rectangle (e.g. A5 aspect), inverse-warp with bilinear sampling; (6) normalize lighting (background subtraction/contrast stretching, adaptive threshold) then OCR. If the page is almost fronto-parallel an affine map (3 corners) suffices.